
November 26, 2025

Quiz-2 problems: Solve by writing codes & showing the results in graphs for problems 1-5 given on pages 1-2.

1. Suppose a student has a meteorology data on rainfall distributions f over some selected months x as follows. (a) $x = \{0, 1, 2, 3, 4\}$, $f_1 = \{5, 9, 18, 25, 51\}$, and (b) $x = \{0, 1, 2, 3, 4\}$, $f_2 = \{1, 1, 9.2, 100, 169\}$. Using Linear and Least squares fit, determine values of f at $x = 1.5$.
2. Determine the integration values:

a)

$$S = \int_0^{\frac{3}{4}\pi} (1 + x + \sin(x)) dx.$$

b)

$$S = \int_0^3 x^2 dx$$

Using Simpson and Trapezoid rules.

3. Determine the differentiation value of

a)

$$f(x) = 1 + x + \sin(x)$$

in the range $x \in [0, \frac{3}{4}\pi]$

b)

$$f(x) = x^2 - 2x + 1$$

in the range $[2.5, 3.5]$.

Use FD, BD, & MD rules to show differences in graphs.

Q4

Curve fitting and Interpolation.

#4. The following measurement from spectroscopy is recorded. With curve fitting approach, explain the global nature of the data.

x (depth of atomic layer)	f(x) (electron scattering intensity)
-0.83836	8.54662
-0.383775	22.7661
0.0358423	39.2307
0.525396	50.0075
1.01495	39.5300
1.18979	31.1480
1.43457	19.1738
1.81921	7.79823
2.37870	1.66143
2.97316	0.164646

#5. A data from meteorology center shows how humidity affects rainfall in arbitrary units.

x (humidity)	f(x) (rainfall)
0.24	1000
0.17	500
0.38	1200
0.46	1600
0.89	1800

What is global nature of the relation between precipitation & rainfall? What is best estimate of rainfall when the precipitation is 0.69?